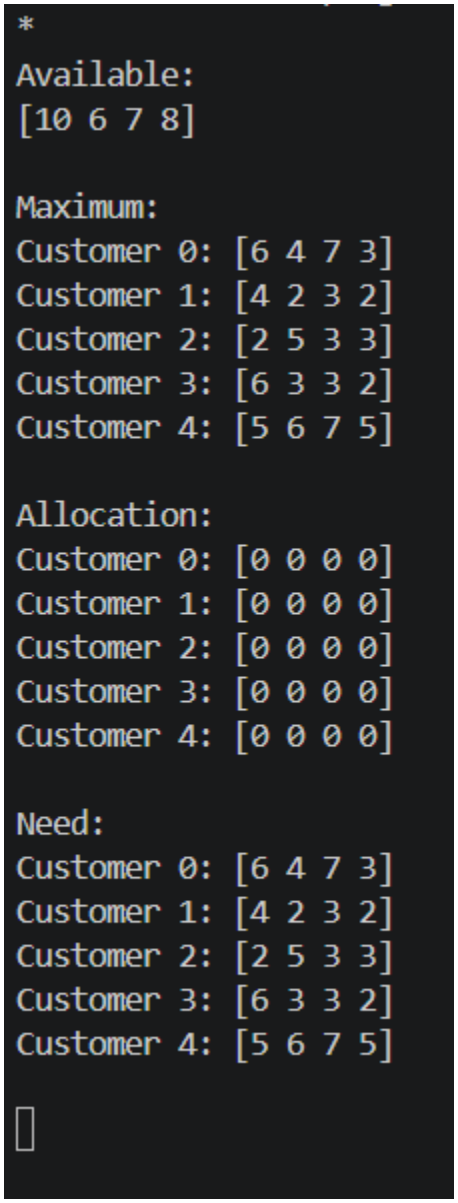
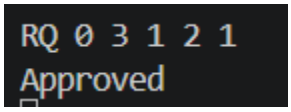
Test 1 Initial State:



Test 2 Valid Safe Request:



```
*
Available:
[7 5 5 7]

Maximum:
Customer 0: [6 4 7 3]
Customer 1: [4 2 3 2]
Customer 2: [2 5 3 3]
Customer 3: [6 3 3 2]
Customer 4: [5 6 7 5]

Allocation:
Customer 0: [3 1 2 1]
Customer 1: [0 0 0 0]
Customer 2: [0 0 0 0]
Customer 3: [0 0 0 0]
Customer 4: [0 0 0 0]

Need:
Customer 0: [3 3 5 2]
Customer 1: [4 2 3 2]
Customer 2: [2 5 3 3]
Customer 3: [6 3 3 2]
Customer 4: [5 6 7 5]
```

Test 3 Request Exceeds Need:

```
RQ 0 4 0 0 0
Denied
```

Test 4 Insufficient Available Resources:

```
RQ 1 8 0 0 0
Denied
```

Test 5 Unsafe Request:

```
RQ 4 5 3 5 4
Denied
```

Test 6 Release Resource:

```
RL 0 1 0 1 0
*
Available:
[8 5 6 7]

Maximum:
Customer 0: [6 4 7 3]
Customer 1: [4 2 3 2]
Customer 2: [2 5 3 3]
Customer 3: [6 3 3 2]
Customer 4: [5 6 7 5]

Allocation:
Customer 0: [2 1 1 1]
Customer 1: [0 0 0 0]
Customer 2: [0 0 0 0]
Customer 3: [0 0 0 0]
Customer 4: [0 0 0 0]

Need:
Customer 0: [4 3 6 2]
Customer 1: [4 2 3 2]
Customer 2: [2 5 3 3]
Customer 3: [6 3 3 2]
Customer 4: [5 6 7 5]

```

Test 7 Invalid Release:

```
RL 0 0 5 0 0
Error: Cannot release more resources than allocated
```

Test 8 Multiple Transactions:

```
 TestBankers inPutFile.txt 10 6 7 8
Commands:
To request resources enter: RQ
To release resources enter: RL
To display the current state enter: *
To terminate the program enter: T
RQ 0 3 1 2 1
Approved
RQ 2 1 2 1 1
Approved
RL 0 1 0 1 0
RQ 3 2 1 1 1
Approved
*
Available:
[5 2 4 5]

Maximum:
Customer 0: [6 4 7 3]
Customer 1: [4 2 3 2]
Customer 2: [2 5 3 3]
Customer 3: [6 3 3 2]
Customer 4: [5 6 7 5]

Allocation:
Customer 0: [2 1 1 1]
Customer 1: [0 0 0 0]
Customer 2: [1 2 1 1]
Customer 3: [2 1 1 1]
Customer 4: [0 0 0 0]

Need:
Customer 0: [4 3 6 2]
Customer 1: [4 2 3 2]
Customer 2: [1 3 2 2]
Customer 3: [4 2 2 1]
Customer 4: [5 6 7 5]
```